Lec.1

Artificial Intelligence

What is Artificial Intelligence?

[Def. 1] **Artificial intelligence (AI) may be defined as the branch of computer science that is concerned with the automation of intelligent behavior**. This definition is particularly appropriate in that it emphasizes a conviction that **AI is a part of computer science** and, as such, **must be based on sound theoretical and applied principles of that field**. These principles include the data structures used in knowledge representation, the algorithms needed to apply that knowledge, and the languages and programming techniques used in their implementation.

However, this definition suffers from the fact that intelligence itself is not very well defined or understood. Although most of us are certain that we know intelligent behavior when we see it, it is doubtful that anyone could come close to defining intelligence in a way that would be specific enough to help in the evaluation of a supposedly intelligent computer program, while still capturing the vitality and complexity of the human mind [Luger 2009].

[Def. 2] **Artificial intelligence (AI) is the field that studies the synthesis and analysis of computational agents that act intelligently**. Let us examine each part of this definition.

An agent is something that acts in an environment; it does something. Agents include worms, dogs, thermostats, airplanes, robots, humans, companies, and countries. We are interested in what an agent does; that is, how it acts. We judge an agent by its actions. **An agent acts intelligently when**

- **What it does is appropriate for its circumstances and its goals, considering the short term and long-term consequences of its actions**
- **It is flexible to changing environments and changing goals**
- **It learns from experience**
- **It makes appropriate choices given its perceptual and computational limitations**

A Computational agent is an agent whose decisions about its actions can be explained in terms of computation. That is, **the decision can be broken down into primitive operations that can be implemented in a physical device**. This computation can take many forms. In humans this computation is carried out in “wetware”; in computers it is carried out in “hardware.” Although there are some agents that are arguably not computational, such as the wind and rain eroding a landscape, it is an open question whether all intelligent agents are computational [Poole and Mackworth].

What can AI do today?

A concise answer is difficult because there are so many activities in so many subfields. Here I sample a few applications;

Robotic vehicles: A driverless robotic car named STANLEY sped through the rough terrain of the Mojave dessert at 22 mph, finishing the 132-mile course first to win the 2005 DARPA Grand Challenge. STANLEY is a Volkswagen Touareg outfitted with cameras, radar, and laser rangefinders to sense the environment and onboard software to command the steering, braking, and acceleration.

Speech recognition: A traveller calling United Airlines to book a flight can have the entire conversation guided by an automated speech recognition and dialog management system.

Autonomous planning and scheduling: A hundred million miles from Earth, NASA's Remote Agent program became the first on-board autonomous planning program to control the scheduling of operations for a spacecraft. REMOTE AGENT such as Mars Exploration Rove (Opportunity) nicknamed Oppy, generated plans from high-level goals specified from the ground and monitored the execution of those plans—detecting, diagnosing, and recovering from problems as they occurred.

Game playing: IBM's DEEP BLUE became the first computer program to defeat the world champion in a chess match when it bested Garry Kasparov by a score of 3.5 to 2.5 in an exhibition match. Kasparov said that he felt a "new kind of intelligence" across the board from

Spam fighting: Each day, learning algorithms classify over a billion messages as spam, saving the recipient from having to waste time deleting what, for many users, could comprise 80% or 90% of all messages, if not classified away by algorithms. Because the spammers are continually updating their tactics, it is difficult for a static programmed approach to keep up, and learning algorithms work best.

Logistics planning: During the Persian Gulf crisis of 1991, U.S. forces deployed a Dynamic Analysis and Replanning Tool, DART, to do automated logistics planning and scheduling for transportation. This involved up to 50,000 vehicles, cargo, and people at a time, and had to account for starting points, destinations, routes, and conflict resolution among all parameters.

The AI planning techniques generated in hours a plan that would have taken weeks with older methods. The Defense Advanced Research Project Agency (DARPA) stated that this single application more than paid back DARPA's 30-year investment in AI

Robotics: The iRobot Corporation has sold over two million Roomba robotic vacuum cleaners for home use. The company also deploys the more rugged PackBot to Iraq and Afghanistan, where it is used to handle hazardous materials, clear explosives, and identify the location of snipers.

Machine Translation: A computer program automatically translates from Arabic to English, allowing an English speaker to see the headline "Ardogan Confirms That Turkey Would Not Accept Any Pressure, Urging Them to Recognize Cyprus." The program uses a statistical model built from examples of Arabic-to-English translations and from examples of English text totalling two trillion words. None of the computer scientists on the team speak Arabic, but they do understand statistics and machine learning algorithms

These are just a few examples of artificial intelligence systems that exist today. Not magic or science fiction—but rather science, engineering, and mathematics. The two most fundamental concerns of AI researchers are knowledge representation and search, to which this course provides an introduction. The first of these addresses the problem of capturing in a language, i.e., one suitable for computer manipulation, the full range of knowledge required for intelligent behaviour. Search is a problem solving technique that systematically explores a space of problem states, i.e., successive and alternative stages in the problem-solving process

Artificial Neural Networks (ANN): is a computational model inspired by the human brain's structure, consisting of interconnected nodes (artificial neurons) organized in layers. ANNs process data by passing signals through these layers, using weights to learn patterns and make predictions or classifications. They are fundamental to machine learning and AI, enabling applications like image recognition and financial forecasting.

Why Neural Networks And Why now?

- As modern computers become ever more powerful, scientists continue to be challenged to use machines effectively for tasks that are relatively simple of humans.
- We learn easily to recognize the letter A or distinguish a cat from a bird.
- More experience allows us to refine our responses and improve our performance.
- Even without a teacher, we can group similar patterns together.

Brains versus Computers : Some numbers

1. There are approximately 10 billion neurons in the human cortex, compared with 10 of thousands of processors in the most powerful parallel computers.
2. Each biological neuron is connected to several thousands of other neurons, similar to the connectivity in powerful parallel computers.
3. Lack of processing units can be compensated by speed. The typical operating speeds of biological neurons is measured in milliseconds (10^{-3} s), while a silicon chip can operate in nanoseconds (10^{-9} s).
4. The human brain is extremely energy efficient, using approximately 10^{-16} joules per operation per second, whereas the best computers today use around 10^{-6} joules per operation per second.
5. Brains have been evolving for tens of millions of years, computers have been evolving for tens of decades.

What is a Neural Net. ?

An artificial neural network is an information-processing system that has certain performance characteristics in common with biological neural networks. Artificial neural networks have been developed as generalizations of mathematical models of human cognition or neural biology, based on the assumptions that:

1. Information processing occurs at many simple elements called neurons.

2. Signals are passed between neurons over connection links.

3. Each connection link has an associated weight, which, in a typical neural net, multiplies the signal transmitted.

4. Each neuron applies an activation function (usually nonlinear) to its net input (sum of weighted input signals) to determine its output signal.

A neural network is characterized by:

(1) its pattern of connections between the neurons (called its **architecture**)

(2) its method of determining the weights on the connections (called its **training**, or **learning**, **algorithm**)

(3) its **activation function**.

key features

- Several key features of the processing elements of artificial neural networks are:
- The processing element receives **many signals**.

- The processing element **sums the weighted** inputs.
- Under appropriate circumstances (sufficient input), the neuron **transmits a single output**.
- The output from a particular neuron may **go to many other neurons** (the axon branches).

Biological Neural Networks

- A biological neuron has three types of components that are of particular interest in understanding an artificial neuron: its **dendrites**, **soma**, and **axon**.
- The many dendrites receive signals from other neurons. The signals are electric impulses that are transmitted across a synaptic gap by means of a **chemical process**.
- The action of the chemical transmitter **modifies the incoming signal** (typically, by scaling the frequency of the signals that are received) in a manner similar to the action of the weights in an artificial neural network.

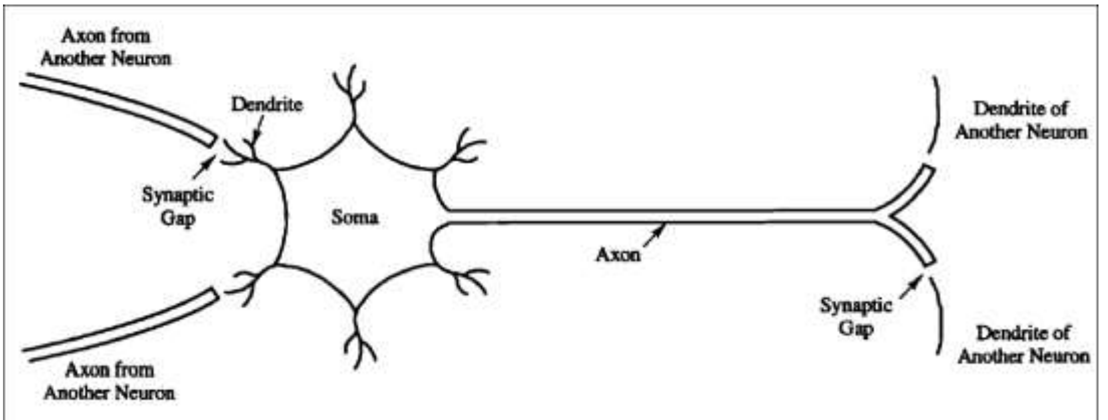


Figure 1.3 Biological neuron.

If-Then Rules

Where Are Neural Nets Being used?

1. Signal Processing.

2. Control.

3. Pattern Recognition.

4. Medicine.

5. Speech Production.

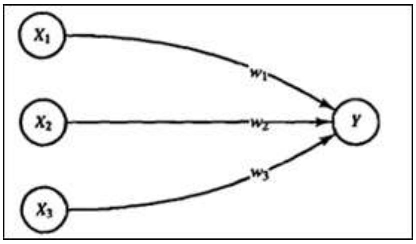
6. Speech Recognition.

7. Businesses

Activation or Activity Level

For example, consider a neuron Y, that receives inputs from neurons X_1, X_2, and X_3 The weights on the connections from X_1, X_2, and X_3 to neuron Y are w_1 to W_2, and W_3, respectively. The net input, y_in, to neuron Y is the sum of the weighted signals from neurons X_1, X_2, and X_3,i.e.

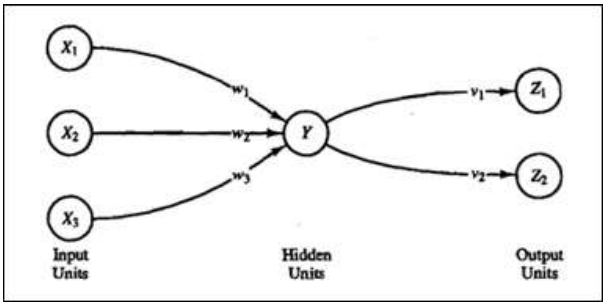
$$y\text{-}in = W_1X_1 + W_2X_2 + W_3X_3$$



$$f(x) = \frac{1}{1+\exp{(-x)'}}$$

The activation y of neuron Y is given by some function of its net input, y = f(y_in), e.g., the logistic sigmoid function (an S-shaped curve)

Now suppose further that neuron Y is connected to neurons Z_1 and Z_2,with weights v_1 and v_2,respectively. Neuron Y sends its signal y to each of these units. However, in general, the values received by neurons Z_1 andZ_2 will be different, because each signal is scaled by the appropriate weight, v_1 or v_2 .In a typical net, the activations Z_1 and Z_2 of neurons Z_1 and Z_2 would depend on inputs from several or even many neurons, not just one, as shown in this simple example.



Common Activation Functions

1. Identity function: f (x) = x for all x

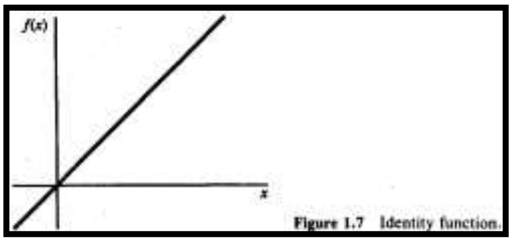


Figure 1.7 Identity function.

f (x) = x for all x

2. Binary step function (with threshold Θ):

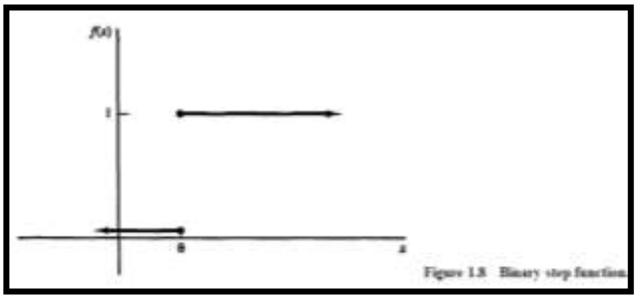


Figure 1.8 Binary step function

$$f(x) = \begin{Bmatrix} 1 \;\; if \;\; x \geq \theta \\ 0 \;\; if \; x \; < \; \theta \end{Bmatrix}$$

3. Binary sigmoid:

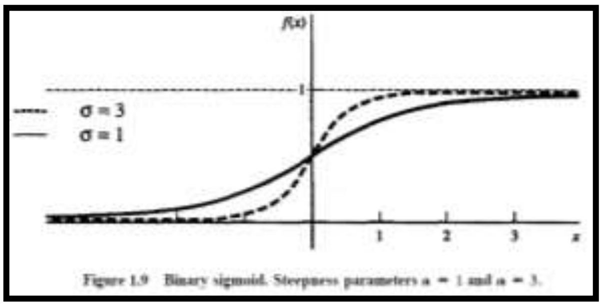
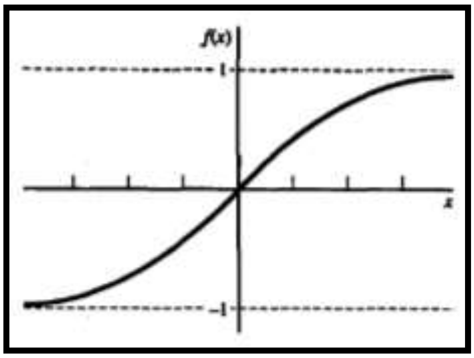


Figure 1.9 Binary sigmoid. Steepness parameters σ = 1 and σ = 3.

$$f(x) = \frac{1}{1 + \exp(-\sigma x)},$$

$$f'^{(x)} = \sigma f(x)[1 - f(x)].$$

4. Bipolar sigmoid:



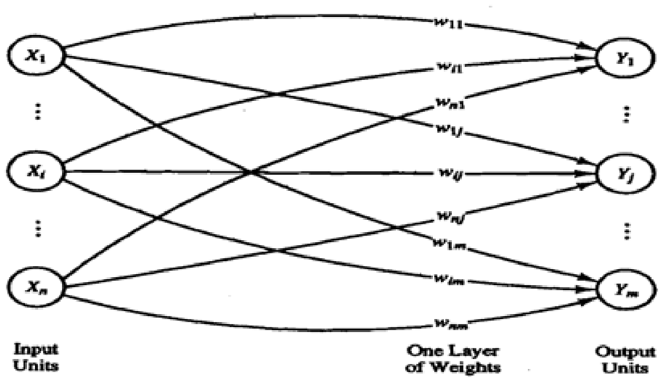
$$g(x) = 2f(x) - 1 = \frac{2}{1 + \exp(-ax)} - 1$$

$$= \frac{1 - \exp(-\sigma x)}{1 + \exp(-\sigma x)}.$$

$$g'(x) = \frac{\sigma}{2}[1 + g(x)][1 - g(x)].$$

Typical Architectures

- The arrangement of neurons into layers and the connection patterns within and between layers is called the **net architecture**.
- **Feed forward nets**: nets in which the signals flow from the input units to the output units, in a forward direction.
- **Recurrent net:** in which there are closed-loop signal paths from a unit back to itself.
- **A single-layer net:** has one layer of connection weights. Often, the units can be distinguished as input units, which receive signals from the outside world, and output units, from which the response of the net can be read.



Setting the Weights

- In addition to the architecture, the method of setting the values of the weights **(training)** is an important distinguishing characteristic of different neural nets.
- The weights are then adjusted according to a learning algorithm. This process is known as **supervised training**.
- **Unsupervised training** Self-organizing neural nets group similar input vectors together with out the use of training data to specify what a typical member of each group looks like or to which group each vector belongs.